# Top-$k$ Identification from Noisy Pairwise Comparisons under Restricted Comparison Graphs

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**Abstract**

We study how restricting which pairs may be compared changes the sample complexity of identifying the top-$k$ items under a Bradley–Terry model. We assume comparisons are only possible along the edges of a given, connected comparison graph $G$ with algebraic connectivity $\lambda_G$. On the algorithmic side, a one-shot estimator that compares every edge a suitable number of times and thresholds a gauge-constrained maximum-likelihood estimate identifies the top-$k$ set with probability $1-\delta$ using $T \le C\, m\, \kappa^2 \log(n/\delta) \,/\, (\sigma_{\min} \lambda_G \Delta^2)$ comparisons, where $m$ is the number of edges, $\Delta$ is the gap between the $k$-th and $(k{+}1)$-th utilities, and $\kappa$ depends only on the dynamic range; in words, the budget grows with the number of comparable pairs, inversely with how well connected the graph is, and quadratically in the inverse gap. On the information-theoretic side we prove a lower bound for the natural class of edge-balanced algorithms that recovers the same $\lambda_G$ and $\Delta$ dependence up to constants and a factor $n$ on bounded-degree graphs, together with a universal two-point bound; whether the factor $n$ can be removed for fully adaptive algorithms is open. For the complete graph both bounds degenerate to the classical full-access rates, as they must. Experiments on complete, random, star, dumbbell and cycle graphs are consistent with the $n$-scaling of the bound for the analyzed estimator, show that the measured gap-dependence of the practical algorithm is softer than the predicted $\Delta^{-2}$ – a deviation we derive from its stopping rule under a stated variance scaling – exhibit the connectivity penalty when the top-$k$ boundary straddles a bottleneck, and reveal a sharp failure mode of the uniform-sampling majority-vote baseline on sparse graphs; they also show that adaptive query concentration can beat the worst-case connectivity dependence.

## 1 Introduction

Pairwise-comparison data underlies ranking systems in which only *relative* feedback is available: model evaluation platforms compare two systems at a time, peer grading compares two submissions, and preference elicitation compares two alternatives. In several of these applications the set of comparable pairs is *restricted* before any data is collected: leaderboards pit same-size systems against each other, tournaments fix pairings in advance, peer grading compares submissions within a track. The feasible pairs form a comparison graph $G$ given to the learner, and each edge may be measured repeatedly.

This paper asks how the connectivity of $G$ affects the number of comparisons needed to identify the top-$k$ items with probability $1-\delta$ under a Bradley–Terry (BT) model. On the complete graph the problem reduces to the classical full-access setting, where the sample complexity is $\Theta\left(\frac{n}{k}\,\frac{\log n}{\Delta^2}\right)$ up to log factors [27, 16, 8, 13]. For a general graph the answer must depend on how easily information travels in $G$: for utility *estimation*, the algebraic connectivity $\lambda_G$ (the second-smallest

Laplacian eigenvalue) is known to govern the difficulty [26, 19, 33, 14]. Our contribution transports this dependence to the *identification* problem, where the object of interest is a decision (the top-$k$ set) rather than the utility vector.

**Contributions.** To the best of our knowledge, no prior work states the sample complexity of $\delta$-correct top-$k$ identification under a fixed, given comparison graph in terms of $\lambda_G$; the closest works either study the full-access (or random-design) problem [27, 8, 13], semi-random comparison graphs [32], or estimation rather than identification on fixed graphs [33, 19]. Our concrete contributions are:

1. **Upper bound (Theorem 5.1).** A one-shot estimator that compares every edge of $G$ the same number of times and thresholds a constrained MLE identifies the top-$k$ set with probability $1-\delta$ using $T \le C\, m_{\text{edges}}\, \kappa^2 \log(n/\delta) \,/\, (\sigma_{\min} \lambda_G \Delta^2)$ comparisons, where $m_{\text{edges}} = |E(G)|$, $\sigma_{\min} = \sigma'(2B)$ and $\kappa$ depends only on the dynamic range $B$. For bounded-degree graphs $m_{\text{edges}} = \Theta(n)$, so $T = \tilde{O}(n \,/\, (\lambda_G \Delta^2))$.

2. **Lower bound (Theorem 4.1).** (a) Every $\delta$-correct algorithm needs $\Omega(\log(1/\delta)/\Delta^2)$ comparisons on any graph. (b) Every $\delta$-correct algorithm that is *edge-balanced* (Definition 3.3) needs $\Omega\big(m \log(1/\delta) \,/\, (n\, \lambda_G\, \Delta^2)\big)$ comparisons. On bounded-degree graphs this captures the same $\lambda_G$ and $\Delta$ dependence as the upper bound but sits a factor $n$ below it, and it holds under the range side condition of the theorem; removing the balancedness restriction and the factor $n$ is open (Conjecture 4.2), and Remark 4.3 explains why the balancedness assumption cannot simply be dropped.

3. **Complete-graph consistency (Proposition 5.4).** Both bounds degenerate to the classical full-access rates, which serves as a self-check of the derivations.

4. **Experiments.** On complete, random, star, dumbbell and cycle graphs (bounded-range instances) we measure the $n/\Delta^2$ proportionality, the $\Delta$-dependence of $T$ over a dense sweep (softer than $\Delta^{-2}$, with the deviation derived from the stopping rule), the role of $\lambda_G$ for the analyzed estimator including a boundary-across-bottleneck dumbbell experiment, the error–budget tradeoff, and a stark failure of the majority-vote baseline on the cycle, where its error stays at or above 0.90 at every budget while the proposed algorithm succeeds.

**Honesty about scope.** Our upper bound is proved for a one-shot uniform-per-edge estimator, while the practical algorithm we evaluate (Algorithm 1) is calibrated-heuristic; the lower bound of Theorem 4.1(b) covers edge-balanced algorithms (Remark 4.3); all reported error rates are measured, never assumed. Section 7 elaborates.

## 2 Related Work

**Top-$k$ and ranking with full access.** For the complete-graph setting, Copeland counting is optimal up to constants without parametric assumptions [27], spectral MLE is minimax under random designs [9, 8], approximate top-$k$ is characterized up to logs [15, 16], the multinomial logit model admits instance-optimal algorithms [6], full ranking is minimax under Kendall's tau [5], noisy sorting rates are known [29], and fixed-confidence top-$k$ is asymptotically optimal [13] — all allowing *any* pair. Multi-way comparisons, inference and sparse-pairwise LLM ranking appear in [12, 4, 18]; an RLHF perspective in [34].

**Estimation under general comparison graphs.** For *estimating* utilities, topology-dependent minimax bounds via the Laplacian spectrum are given by [26], $\ell_\infty$ MLE bounds in terms of the algebraic connectivity by [19], entrywise guarantees of order $1/\sqrt{\lambda_{\mathcal{D}}}$ on arbitrary fixed graphs by [33], effective-resistance characterizations by [7, 14], and Rank Centrality on random graphs by [21]; informative designs are well-connected graphs [22]. Our upper bound is a corollary of [33] specialized to the uniform design; the identification question and its $\delta$-correct sample complexity are what we add.

**Graph-constrained identification.** Closest to us, [32] studies top-$K$ ranking under a *semi-random* graph with near-optimal sample complexity up to $\log^2(n)$; the graph is not fixed and no $\lambda_G$-explicit bound is given. Pure exploration under *feedback graphs* [23, 10] constrains which *observations* follow one arm pull, unlike our constraint on which *pairs* may be compared; structured pure exploration also appears in [25, 2].

**Dueling bandits.** This literature mainly studies regret [35, 1, 30, 20, 11, 28, 31, 24]; identification-type results exist for Copeland winners with full access [3], and early active ranking under structural assumptions in [17]. None restricts comparisons to the edges of a given graph.

## 3 Problem Setup

**Items and utilities.** Let $[n] = \{1, \dots, n\}$ be a set of items, each item $i$ having an unknown utility $\theta_i \in \mathbb{R}$.

**Comparison graph.** Let $G = ([n], E)$ be a connected undirected graph; a comparison between $i$ and $j$ is possible only when $\{i, j\} \in E$. We write $m = |E|$ for the number of edges, $L_G$ for the graph Laplacian, and $\lambda_G > 0$ for its second-smallest eigenvalue (the algebraic connectivity). For a vector $w \in \mathbb{R}^E$ of nonnegative edge weights we write $L_w = \sum_{e=(i,j)\in E} w_e \, (e_i - e_j)(e_i - e_j)^\top$ for the weighted Laplacian.

**Bradley–Terry model.** A comparison of $\{i, j\}$ produces the winner $Y \in \{i, j\}$ with

$$\Pr(Y = i) = \sigma(\theta_i - \theta_j), \qquad \sigma(x) = \frac{1}{1 + e^{-x}}. \tag{1}$$

**Gap.** Let $\theta_{(1)} \geq \theta_{(2)} \geq \cdots \geq \theta_{(n)}$ be the sorted utilities and let $S^\star = \{(1), \dots, (k)\}$ be the true top-$k$ set. The *gap* is

$$\Delta = \theta_{(k)} - \theta_{(k+1)} > 0. \tag{2}$$

**Goal.** An algorithm observes $G$, adaptively selects edges of $G$ (possibly repeatedly), observes BT outcomes, and outputs $\hat{S} \subseteq [n]$ with $|\hat{S}| = k$. It is $\delta$-*correct* if $\Pr(\hat{S} = S^\star) \geq 1 - \delta$ for every utility vector. $T^\star(n, k, \Delta, G, \delta)$ denotes the worst-case, over utility vectors, of the expected total number of comparisons used by a $\delta$-correct algorithm (the expectation is over the comparison outcomes and the algorithm's randomness); our lower bounds bound this expectation, and our experiments report the empirical distribution of $T$.

**Bounded dynamic range.** Several statements below use the standard bounded-range condition.

**Condition 3.1** (Bounded range)**.** $\|\theta\|_\infty \le B$ for a known $B$, and $\sigma_{\min} := \sigma'(2B) > 0$.

For two instances $\theta, \theta'$ and an edge $e = (i, j)$ let $h_e = (\theta_i - \theta_j) - (\theta'_i - \theta'_j)$ denote the change of the edge's log-odds. The single-comparison Kullback–Leibler divergence obeys the following bound (chi-square domination under a bounded density ratio).

**Lemma 3.2** (Per-comparison KL)**.** *If both instances satisfy Condition 3.1 with range $B$, then for every edge $e$,*

$$\mathrm{KL}\big(\mathrm{Ber}(\sigma(\theta_i - \theta_j)) \,\big\|\, \mathrm{Ber}(\sigma(\theta'_i - \theta'_j))\big) \;\le\; C_B\, h_e^2, \qquad C_B = \frac{1}{16\,\sigma(-2B)\,\big(1 - \sigma(-2B)\big)}.$$

*Proof.* Write $p = \sigma(x)$, $q = \sigma(x+h)$ with $|x|, |x+h| \le 2B$; then $p, q \in [p_0, 1-p_0]$ with $p_0 = \sigma(-2B)$. The chi-square divergence dominates the KL divergence, and for Bernoulli laws $\chi^2(p\|q) = \frac{(p-q)^2}{q} + \frac{(p-q)^2}{1-q} = \frac{(p-q)^2}{q(1-q)}$. Since $|p - q| = |\sigma(x+h) - \sigma(x)| \le \frac{1}{4}|h|$ and $q(1-q) \ge p_0(1-p_0)$ on the range, we obtain $\mathrm{KL} \le \frac{h^2}{16\,p_0(1-p_0)} = C_B\, h^2$. □

**Definition 3.3** (Edge-balanced algorithm)**.** An algorithm is *$\alpha$-balanced* on $G$ if for every instance $\theta$ and every edge $e \in E$, $\mathbb{E}_\theta[m_e] \;\le\; \frac{\alpha}{m}\,\mathbb{E}_\theta[T]$, where $m_e$ is the number of comparisons placed on edge $e$, $T$ the total number of comparisons, and $\alpha \ge 1$ a constant. The uniform-per-edge estimator of Section 5 has $\alpha = 1$; a random walk on a regular graph mixed to stationarity has $\alpha = O(1)$.

# 4 Lower Bound

**Theorem 4.1** (Lower bound)**.** *Fix a connected $G = ([n], E)$ with $m$ edges, $1 \le k < n$ and gap $\Delta > 0$. Then:*

*(a) every $\delta$-correct algorithm needs $T \;\ge\; \dfrac{\log(1/(4\delta))}{4\,C_B\,\Delta^2}$ comparisons, regardless of balancedness;*

*(b) if moreover Condition 3.1 holds with a range $B \ge 2\sqrt{n}\,\Delta$ (a sufficient condition), then every $\delta$-correct $\alpha$-balanced algorithm needs $T \;\ge\; \dfrac{m}{\alpha\,C_B\,n\,\lambda_G\,\Delta^2}\,\log\frac{1}{4\delta}$.*

*Proof.* **(a) Two-point construction at the boundary.** Let $a$ be the item of rank $k$ and $b$ that of rank $k+1$ under the instance $\theta$, which we design with $\theta_a - \theta_b = \Delta$ and with every other item at distance at least $3\Delta$ from both $\theta_a$ and $\theta_b$; such an instance exists for any $\Delta$ and any desired positions of $a, b$ in the graph. Let $\theta' = \theta + \Delta(e_b - e_a)$, so that $\theta'_b - \theta'_a = \Delta$ and the top-$k$ sets of $\theta$ and $\theta'$ differ in exactly one element; every other item moves by at most $\Delta$ and cannot cross the boundary by the distance condition. For any edge $e = (i, j)$ not incident to $a$ or $b$ we have $h_e = 0$, and for edges incident to $a$ or $b$, $|h_e| \le 2\Delta$. Let $\mathcal{A}$ be any algorithm and let $N_{ab}$ be the number of comparisons it places on edges incident to $a$ or $b$; then $N_{ab} \le T$. By the chain rule and Lemma 3.2,

$$\mathrm{KL}\Big(P_\theta^{\mathcal{A}} \,\Big\|\, P_{\theta'}^{\mathcal{A}}\Big) \;\le\; \mathbb{E}_\theta[\textstyle\sum_t \mathrm{KL}_t] \;\le\; 4C_B\Delta^2\,\mathbb{E}_\theta[N_{ab}] \;\le\; 4C_B\Delta^2\,T.$$

Bretagnolle–Huber in its mixture form bounds the Bayes error from below by $\frac{1}{4}e^{-\mathrm{KL}}$; if $\mathcal{A}$ is $\delta$-correct on both instances, its error under the equal-weight mixture is at most $\delta$, hence $\mathrm{KL} \ge \log(1/(4\delta))$, and $T \ge \log(1/(4\delta))/(4C_B\Delta^2)$.

**(b) Fiedler-perturbation construction.** Let $u$ be a unit-norm eigenvector of $L_G$ for $\lambda_G$, so that $u \perp \mathbf{1}$ and $\sum_{e=(i,j)}(u_i - u_j)^2 = \lambda_G$. The entries of $u$ then have variance $1/n$, and the Popoviciu inequality $(\max_i u_i - \min_i u_i)^2 \geq 4\,\mathrm{Var}(u)$ gives, for $a = \arg\max_i u_i$ and $b = \arg\min_i u_i$, $(u_a - u_b)^2 \geq 4/n$. Design $\theta$ (possible by the freedom in choosing utilities away from the boundary) such that: (i) $\theta_a - \theta_b = \Delta$, with $a$ of rank $k$ and $b$ of rank $k+1$; (ii) every other item lies at distance at least $2\varepsilon + \Delta$ from both $\theta_a$ and $\theta_b$, where $\varepsilon = 2\Delta/(u_a - u_b) \leq \Delta\sqrt{n}$; (iii) $\|\theta\|_\infty \leq B$, which is compatible with (ii) under the sufficient hypothesis $B \geq 2\sqrt{n}\,\Delta$. Let $\theta' = \theta - \varepsilon u$. Then $\theta'_a - \theta'_b = \theta_a - \theta_b - \varepsilon(u_a - u_b) = -\Delta$, so the top-$k$ set changes by one element, while every other item moves by at most $\varepsilon\|u\|_\infty \leq \varepsilon$ and cannot cross the boundary by (ii). For an edge $e = (i,j)$ the log-odds change is $h_e = -\varepsilon(u_i - u_j)$. For an $\alpha$-balanced algorithm, $\mathbb{E}_\theta[m_e] \leq \frac{\alpha}{m}\,\mathbb{E}_\theta[T]$ for every edge $e$, so by Lemma 3.2 and the chain rule,

$$\mathrm{KL}\Big(P_\theta^{\mathcal{A}} \,\Big\|\, P_{\theta'}^{\mathcal{A}}\Big) \;\leq\; C_B\varepsilon^2 \sum_{e\in E} \mathbb{E}[m_e]\,(u_i - u_j)^2 \;\leq\; \frac{\alpha T}{m}\, C_B\varepsilon^2 \sum_{e} (u_i - u_j)^2 \;=\; \frac{\alpha T}{m}\, C_B\varepsilon^2\, \lambda_G.$$

Bretagnolle–Huber as in part (a) forces $\mathrm{KL} \geq \log(1/(4\delta))$, and substituting $\varepsilon = 2\Delta/(u_a - u_b)$ with $(u_a - u_b)^2 \geq 4/n$ yields $T \;\geq\; \dfrac{m\,(u_a - u_b)^2}{4\alpha C_B \lambda_G \Delta^2} \log\frac{1}{4\delta} \;\geq\; \dfrac{m}{\alpha C_B\, n\, \lambda_G\, \Delta^2} \log\frac{1}{4\delta}$. □

**Conjecture 4.2** (Fully adaptive rate)**.** *There is a constant $c > 0$ such that every $\delta$-correct algorithm (no balancedness restriction) on a bounded-degree graph with algebraic connectivity $\lambda_G$ needs $T \geq c\,\frac{n}{\lambda_G\,\Delta^2}\log\frac{1}{\delta}$ comparisons for some instance satisfying Condition 3.1. We do not know how to prove this and flag the obstruction below.*

**Remark 4.3** (Why balancedness cannot simply be dropped)**.** The proof of Theorem 4.1(b) extends verbatim to *every* algorithm on any graph whose Fiedler edge energy is balanced, in the sense that the concentration ratio $m\,\max_e(u_i - u_j)^2/\lambda_G$ is $O(1)$: the chain-rule bound $\mathrm{KL} \leq C_B\varepsilon^2 \sum_e m_e(u_i - u_j)^2$ is maximized by concentrating queries on the highest-energy edges, and on such graphs even this worst case is within a constant of the balanced bound. On a path graph the energy is nearly uniform – the largest edge difference of the normalized Fiedler vector is $\Theta(n^{-3/2})$, within a constant factor of $\lambda_G/m$ – so the $\lambda_G$ lower bound already covers all algorithms there. On graphs whose Fiedler energy concentrates on few edges, such as two cliques joined by a bridge, an adaptive algorithm can concentrate queries on those edges and the argument degenerates; whether any graph-dependent lower bound survives for fully adaptive algorithms on such graphs is exactly the content of Conjecture 4.2, and our E2/E2c experiments suggest the practical penalty is at most a constant (Section 6).

## 5 Algorithm and Upper Bound

**One-shot estimator (analyzed).** Given $m$ (samples per edge), the estimator UE-TOPK compares every edge $e$ of $G$ exactly $m$ times, fits the gauge-constrained BT maximum-likelihood estimate

$$\hat\theta = \arg\min_{\theta:\ \sum_i \theta_i = 0} \sum_{e=(i,j)\in E} \sum_{\ell=1}^{m} -\log\sigma\big(y_{e,\ell}\,(\theta_i - \theta_j)\big), \tag{3}$$

where $y_{e,\ell} \in \{\pm 1\}$ encodes the winner oriented from the lower-indexed endpoint, and outputs the $k$ largest coordinates of $\hat\theta$. It uses $T = m \cdot m_{\text{edges}}$ comparisons in total.

**Theorem 5.1** (Upper bound)**.** *Let $G$ be connected with $m_{\text{edges}}$ edges and algebraic connectivity $\lambda_G$, suppose Condition 3.1 holds, and assume the* entrywise guarantee*: there exist constants $C$ and $\kappa_0(B) \geq 1$, the latter depending only on $B$, such that the constrained MLE (3), run on any connected weighted graph whose every edge carries at least $m \geq 1$ comparisons, satisfies with probability at least $1 - \delta$*

$$\|\hat{\theta} - \theta\|_\infty \;\leq\; C\,\kappa_0 \sqrt{\frac{\log(n/\delta)}{\sigma_{\min}\, m\, \lambda_G}}.$$

*This guarantee is the specialization to the uniform design of the entrywise analyses of the constrained BT MLE and Rank Centrality [33, 19]; we assume it explicitly because those papers impose regularity assumptions whose verification for arbitrary deterministic graphs we have not carried out here. Under the guarantee there is a universal constant $C_0$ such that: if*

$$m \;\geq\; C_0\, \frac{\kappa_0^2\, \log(n/\delta)}{\sigma_{\min}\, \lambda_G\, \Delta^2},$$

*then* UE-TOPK *is $\delta$-correct. Consequently*

$$T^\star(n, k, \Delta, G, \delta) \;\leq\; T_{\text{UE-TOPK}} \;=\; C_0\, \frac{\kappa^2\, m_{\text{edges}}}{\sigma_{\min}\, \lambda_G\, \Delta^2}\, \log \frac{n}{\delta},$$

*and the bound is achieved by an $\alpha$-balanced algorithm with $\alpha = 1$.*

*Proof.* The design places exactly $m$ comparisons on every edge, so the count-weighted Laplacian is $m\, L_G$ with algebraic connectivity $m\lambda_G$. By the entrywise guarantee – whose specialization to the uniform design follows from the entrywise analyses of [33, 19] – with probability at least $1 - \delta$,

$$\|\hat{\theta} - \theta\|_\infty \;\leq\; C\,\kappa \sqrt{\frac{\log(n/\delta)}{\sigma_{\min}\, m\, \lambda_G}},$$

where the dynamic-range factor $\kappa_0$ accounts for the curvature of $\sigma$ on $[-2B, 2B]$ (in the notation of [33], $\lambda_{\mathcal{D}} = m\lambda_G$ since the design is uniform). For completeness we recall why the $\lambda_G$ appears and no more: the Fisher information of the MLE dominates $\sigma_{\min}\, m\, L_G$ on the mean-zero subspace, so the variance of any contrast $\theta_i - \theta_j$ is at most

$$\frac{2}{\sigma_{\min}\, m}\, (e_i - e_j)^\top L_G^+ (e_i - e_j) \;\leq\; \frac{4}{\sigma_{\min}\, m\, \lambda_G},$$

because for unit-norm contrast vectors $v \perp \mathbf{1}$ one has $v^\top L_G^+ v \leq 1/\lambda_G$ and $\|e_i - e_j\|_2^2 = 2$. Now choose $m$ so that $C\kappa_0\sqrt{\log(n/\delta)/(\sigma_{\min} m \lambda_G)} < \Delta/2$; this is exactly the stated condition with $C_0 = 4C^2$ (the error bound is linear in $\kappa_0$ while the threshold $\Delta/2$ is constant, so solving for $m$ squares $\kappa_0$). On the event $\|\hat{\theta} - \theta\|_\infty < \Delta/2$, every item of $S^\star$ has $\hat{\theta}_i > \theta_{(k)} - \Delta/2$ and every item outside has $\hat{\theta}_j < \theta_{(k+1)} + \Delta/2 < \theta_{(k)} - \Delta/2$, so the $k$ largest coordinates of $\hat{\theta}$ are exactly $S^\star$. The estimator places $m$ samples per edge, hence is 1-balanced, and uses $m \cdot m_{\text{edges}}$ comparisons. □

**Proposition 5.2** (Sequential variant)**.** *Assume Condition 3.1 and the entrywise guarantee of Theorem 5.1 and define, for round $r$ with per-edge budget $c_r$, the data-independent radius $\rho_r := C\kappa_0\sqrt{\log(8n^2 r/\delta)/(\sigma_{\min}\, c_r\, \lambda_G)}$. Run doubling rounds $c_r = 2^r$: each round, fit $\hat{\theta}$ on all data collected so far and eliminate every active item $j$ that is dominated by at least $k$ other active items $i$ in the sense $\hat{\theta}_i - \hat{\theta}_j > 2\rho_r$; stop when $k$ items remain. Then with probability at least $1 - \delta$ the algorithm returns $S^\star$, and it uses at most $2\, m_{\text{edges}}\, \tilde{m}$ comparisons, where $\tilde{m} = 4C^2\kappa_0^2 \log(8n^2 r^\star/\delta)/(\sigma_{\min}\lambda_G\Delta^2)$ equals $m^\star$ of Theorem 5.1 up to log factors ($r^\star$ is the final round).*

**Algorithm 1** GraphWalk-SE (practical variant evaluated in Section 6)

**input:** graph $G$, target $k$, confidence $\delta$; active set $A \leftarrow [n]$; round budget $c \leftarrow 1$; data $\mathcal{D} \leftarrow \emptyset$
**while** $|A| > k$ **do**
    build per-vertex opponent lists: active $v$ samples its *active* $G$-neighbors when it has any, otherwise its full $G$-neighborhood (fallback); with probability 0.1 sample from the full neighborhood (mixing)
    every $v \in A$ performs $c$ comparisons by a random walk from $v$ over these lists; append outcomes to $\mathcal{D}$
    fit the constrained MLE $\hat{\theta}$ on $\mathcal{D}$ (3), with per-edge curvature clipped to $|d| \leq 3$ and a Bernstein linear term in the radii
    compute $P = (H)^+$ from the observed Fisher information; eliminate every $j \in A$ dominated by at least $k$ active items under the contrast screen of Proposition 5.2
    $A \leftarrow$ survivors; $c \leftarrow 2c$
**end while**
**return** the $k$ survivors ordered by $\hat{\theta}$

*Proof.* Apply the entrywise guarantee round by round with a union bound over rounds (per-round confidence $\delta/(2r^2)$ sums to less than $\delta$): on an event of probability at least $1-\delta$, every round-$r$ fit satisfies $\|\hat{\theta}^{(r)} - \theta\|_\infty \leq \rho_r$. On that event, if $i \in S^\star$ and $j \notin S^\star$ are both active then $\theta_i - \theta_j \geq \Delta$, so $\hat{\theta}_i - \hat{\theta}_j \geq \Delta - 2\rho_r > 2\rho_r$ once $\rho_r < \Delta/4$, i.e. once $c_r \geq \tilde{m}$; every non-member is therefore eliminated no later than the first round with $c_r \geq \tilde{m}$. Conversely, suppose inductively that no member of $S^\star$ has been eliminated; then all of $S^\star$ is active, a non-member $i$ satisfies $\hat{\theta}_i - \hat{\theta}_j \leq -\Delta + 2\rho_r < 2\rho_r$ against every member $j \in S^\star$, and only $k-1$ other members exist, so no member is dominated by $k$ actives. The total budget is the geometric sum $\sum_{r \leq r^\star} m_{\text{edges}} 2^r \leq 2 m_{\text{edges}} 2^{r^\star} = 2 m_{\text{edges}} \tilde{m}$. □

**Remark 5.3.** Algorithm 1 below uses a different, calibrated screen: a plug-in Fisher radius with a Bernstein linear term (Appendix A). That screen is a heuristic chosen for empirical efficiency; Proposition 5.2's data-independent screen is the one tied to the entrywise guarantee in Theorem 5.1.

**Proposition 5.4** (Complete-graph degeneration)**.** *Let $G = K_n$. Then $\lambda_G = n$ and $m_{\text{edges}} = n(n-1)/2$, so Theorem 5.1 gives $T \leq C\,\kappa^2\, n \log(n/\delta)/(\sigma_{\min}\Delta^2)$, which is the classical full-access rate up to the factor $k$ (the optimal rate is $\Theta(\frac{n}{k}\,\text{polylog}(n)/\Delta^2)$ [27, 16, 8, 13]); Theorem 4.1(a) gives the matching-in-$\Delta$ two-point lower bound $\Omega(\log(1/\delta)/\Delta^2)$, and Theorem 4.1(b) degenerates to the same order since $m_{\text{edges}}/(n\lambda_G) = (n-1)/(2n) \to \frac{1}{2}$. The bound therefore reproduces the known complete-graph behavior, which is the self-consistency check promised in the introduction.*

## 6 Experiments

All experiments use Bradley–Terry outcomes on bounded-range instances satisfying Condition 3.1 with $B = 3$ (exact instance construction, seed policy and a parameter table are in the supplementary appendix). The practical algorithm is Algorithm 1 with $\delta = 0.05$ and $k = 3$ unless stated otherwise; repetition counts are stated per experiment below. Reported $T$ means carry normal-approximation 95% half-widths (the $T$ distributions are right-skewed, so these are indicative); success and error rates are point estimates whose binomial standard error near the extremes is a few percentage points at our repetition counts, so rate differences below a few points are treated as noise. Subsections are ordered by narrative rather than by experiment number. $\lambda_G$ is computed numerically (NumPy

Table 1: E1: complete graph, $k = 3$, $\delta = 0.05$; mean comparisons over 100 repetitions with 95% half-widths.

| $n$ | $\Delta$ | success | $T$ (mean) | 95% CI | $T\,/\,(n/\Delta^2)$ |
|---|---|---|---|---|---|
| 10 | 0.2 | 1.00 | 20753 | ±1357 | 83 |
| 10 | 0.5 | 1.00 | 5772 | ±221 | 144 |
| 10 | 1.0 | 1.00 | 3140 | ±94 | 314 |
| 20 | 0.2 | 1.00 | 30605 | ±1870 | 61 |
| 20 | 0.5 | 1.00 | 9750 | ±407 | 122 |
| 20 | 1.0 | 1.00 | 6414 | ±259 | 321 |
| 50 | 0.2 | 1.00 | 59083 | ±3035 | 47 |
| 50 | 0.5 | 1.00 | 23379 | ±761 | 117 |
| 50 | 1.0 | 1.00 | 16179 | ±396 | 324 |

`eigvalsh` on the actual Laplacian), never from closed forms. Machine-readable results with all raw per-repetition values are in `results/*.json`; every number quoted below is greppable there.

### 6.1 E1: proportionality to $n/\Delta^2$ on the complete graph

Table 1 reports $T$ for $n \in \{10, 20, 50\}$ and $\Delta \in \{0.2, 0.5, 1.0\}$.

All nine configurations reach empirical success 1.00. The ratio $T/(n/\Delta^2)$ ranges from 47 to 324: it grows with $\Delta$ and decreases in $n$ at fixed $\Delta$; at fixed $\Delta = 0.2$, $T$ grows from 20753 at $n = 10$ to 59083 at $n = 50$ — sublinear in $n$, consistent with the $O(n \log(n/\delta)/\Delta^2)$ upper bound and with early eliminations shrinking the active set. For the *analyzed* estimator we additionally measured the $n$-scaling directly (E2d, Section 6.4): on the complete graph at $\Delta = 0.5$, UE-topk's minimal budget $T^\star$ grows from 720 at $n = 10$ to 3040 at $n = 20$ and 9800 at $n = 50$ (log–log slope 1.60 against the predicted 1 plus a $\log(n/\delta)$ factor, with $m^\star$ quantized on a doubling grid).

### 6.2 E3: the exponent of $\Delta$

Fixing $n = 20$ (complete graph, $k = 3$) we sweep $\Delta$ over 24 log-uniform points in $[0.05, 2.0]$ with 200 repetitions per point, measuring $T$ exactly as throughout (the natural consumption of the $\delta = 0.05$ algorithm; no budget search, no redefinition of $\Delta$). Figure 1 shows the result. Piecewise log–log fits give slope **-1.70** ($R^2 = \mathbf{1.00}$) on $\Delta < 0.2$, **-1.20** on $0.2 \le \Delta \le 0.6$, and **-0.55** on $\Delta > 0.6$: the exponent glides from near $-2$ toward $-1$ as $\Delta$ grows, and even at $\Delta < 0.2$ it does not reach $-2$.

The deviation has a source we can write down: it is encoded in the algorithm's stopping rule. The elimination threshold of Algorithm 1 is $t_{ij} = \sqrt{2L\,s_{ij}^2} + 2L\,s_{ij}^2$ with $L = 2\log(4n^2r/\delta)$ (supplementary appendix); with the empirically measured inverse-counts scaling $s_{ij}^2 = \tilde{\kappa}/c$ – an assumption we state explicitly – the stopping condition $t_{ij} = \Delta$ inverts to the one-parameter law

$$c^\star(\Delta) \;=\; \frac{8A}{\left(\sqrt{1+4\Delta}-1\right)^2}, \qquad A = L\,\tilde{\kappa},$$

whose local log–log exponent glides from $-2$ (as $\Delta \to 0$) to $-1$ (large $\Delta$). Fitting this prediction to the 24 measured means gives $A = 3013.5$ with $R^2 = 0.99$. Equivalently, the two-term model $T = a\,\Delta^{-2} + b\,\Delta^{-1}$ fits with $a = 629.8 \pm 16.4$ and $b = 3430.4 \pm 249.3$ ($R^2 = 0.999$ in raw $T$ space, bootstrap 95% CIs over per-repetition resampling; in log space $R^2 = 0.952$). Three caveats keep this from being a proof of attribution. First, the fits live in raw $T$ space, where the small-$\Delta$ points dominate. Second, at $\Delta = 2.0$ the measured mean is 4180 while the two-term model

Table 2: E2: Algorithm 1 on five graph families ($n = 50$, $\Delta = 0.5$, $k = 3$, 100 repetitions).

| graph | $m_{\text{edges}}$ | $\lambda_G$ | success | $T$ (mean) | rounds (median) |
|---|---|---|---|---|---|
| complete | 1225 | 50.0000 | 1.00 | 23283 | 11 |
| random ($p$=0.15) | 190 | 1.6950 | 1.00 | 286559 | 16 |
| star | 49 | 1.0000 | 1.00 | 39336 | 12 |
| dumbbell | 601 | 0.0743 | 1.00 | 332906 | 13 |
| cycle | 50 | 0.0158 | 1.00 | 57286 | 11 |

predicts 1873 and the one-parameter law 753 (predictions recomputed from the stored coefficients, results/e3_fine.json), so neither model captures the large-$\Delta$ tail, where the round count plateaus at 9 and $T$ flattens into a floor regime both fits omit; the fitted $b/a = 5.45$ also differs from the first-order value 2, indicating that the two-term coefficients absorb effects outside the model. Third, the premises $s_{ij}^2 = \tilde{\kappa}/c$ and a $\Delta$-independent $L$ are approximations. The $\Delta^{-1}$ component is the threshold's linear Bernstein term acting at the decision margin; we read the law as a derived, assumption-explicit description of the small-to-moderate-$\Delta$ regime, not as a proved attribution. We also swept the estimator analyzed in Theorem 5.1 (UE-TOPK) over an 8-point log-uniform subrange of the same $\Delta$ range at 50 repetitions per point: its minimal per-edge budget $m^\star$ is too noisy to resolve an exponent (full-range slope **-1.60**, $R^2 = 0.94$; excluding floor-saturated points with $m^\star \geq 4$, slope **-1.39**, $R^2 = 0.86$, with floor saturation $m^\star \in \{1, 2\}$ at large $\Delta$).
**Verdict.** At measurable gaps *neither* algorithm exhibits the $\Delta^{-2}$ dependence of Theorem 5.1; see Section 6.7(iii).

### 6.3 E2 and E2b: the role of graph connectivity

Table 2 measures the practical algorithm (Algorithm 1) on five graph families at $n = 50$, $\Delta = 0.5$.

All five families reach success 1.00, including the sparse random graph and both bottleneck topologies. The cost, however, does not follow $1/\lambda_G$: a log–log regression of $T$ on $1/\lambda_G$ across the families has slope 0.16. The dumbbell ($\lambda_G = 0.0743$, 332906 comparisons) and the random graph ($\lambda_G = 1.6950$, 286559) are the two most expensive families while the cycle ($\lambda_G = 0.0158$) is among the cheapest despite the smallest $\lambda_G$: active-set concentration makes the practical cost a function of boundary geometry and the active subgraph, not of worst-case connectivity (cf. Remark 4.3); E2b/E2c isolate the connectivity cost for the analyzed estimator.

Theorem 5.1 concerns the *uniform-per-edge* estimator, so we measure its scaling law directly: E2b sweeps the per-edge budget $m$ for UE-TOPK on the five families ($n = 20$, $\Delta = 0.5$, 50 repetitions per point, stopping at the first $m$ with success $\geq 0.95$) and reports $T^\star = m^\star m_{\text{edges}}$. Across the five families the $(m_{\text{edges}}/\lambda_G,\ T^\star)$ pairs are complete (9.5, 1520), star (19, 1216), random (76.8, 4992), cycle (204.3, 640), dumbbell (538.3, 1456), and the log–log slope of $T^\star$ against $m_{\text{edges}}/\lambda_G$ is **-0.06** (the theorem predicts slope 1). We read this as indicative, not as a falsification of the theorem: each $m^\star$ is a threshold estimate from 50 repetitions on a doubling grid (up to a factor-2 quantization), the five-point regression carries no confidence intervals, and neighboring grid decisions are unstable (Section 6.4).

### 6.4 E2c: the connectivity cost made visible

The loose fit of the E2b regression has a geometric reason: the top-$k$ boundary in those instances sits between graph-local items, so information never routes across the bottleneck. E2c forces the issue by placing the boundary *across* the dumbbell's bridge ($k = n/2$): the complete graph reaches




Figure 1: Left: E3, $T$ against $\Delta$ ($n$=20, complete, $k$=3, 24 points × 200 reps), with both fits; error bars not drawn (half-widths in results/e3_fine.json). Right: E4, error rate against the budget cap.

95% success at $T^\star = 9800$ ($m^\star = 8$), the dumbbell needs $T^\star = 38464$ ($m^\star = 64$), a factor 3.92 penalty at identical $(n, k, \Delta)$. The two graphs' $\lambda_G$ (complete 50.0, dumbbell 0.0743) differ by two orders of magnitude, and the bound's $m_{\text{edges}}/\lambda_G$ ratio predicts a factor of about 330 between them (computed in results/e2c_.json), so the connectivity term is directionally correct but loose in constants here; closing the gap via adversarial placements is left open.

### 6.5 E4: error rate under a capped budget

Figure 1 (right) fixes $n = 20$ (complete graph), $\Delta = 0.5$ and caps the comparison budget of Algorithm 1 at $B_{\text{cap}} \in [1000, 16000]$ (a capped run finishes its current doubling round, so realized $T$ can exceed the cap by up to one round); when the budget is exhausted the current MAP estimate is returned. The error rate falls from **0.09** at $B_{\text{cap}} = 1000$ to **0.02** at $B_{\text{cap}} = 2000$ and stays at **0.00** from $B_{\text{cap}} = 3000$ onward, with the uncapped consumption of the algorithm at $T \approx$ **10057**.

### 6.6 E5: the uniform-sampling majority-vote baseline

E5 compares the standard baseline (sample uniform random edges for a fixed budget, rank by win rate) against Algorithm 1 at $n = 20$, $\Delta = 0.5$.

On the complete graph the baseline is *cheaper*: it first reaches an error at or below 0.05 at budget 4000 (measured error 0.01), a factor 0.39 below our algorithm's 10153 — expected, since on $K_n$ the win rate is a faithful Borda proxy while our calibrated radii buy worst-case robustness at a constant-factor price. On the cycle the picture inverts: the baseline's error never drops below 0.90 at any budget up to 128000, because every interior item compares only against its two neighbors and shares approximately the same expected win rate, so win-rate ranking is essentially arbitrary; Algorithm 1 succeeds with 13039 comparisons. Graph-aware estimation, not more sampling, is what restricted graphs demand.

### 6.7 What the experiments do and do not show

Four observations are recorded against the paper's own claims. **(i) Bounded range matters.** All experiments use bounded-range instances (Condition 3.1, $B = 3$); when we additionally ran unbounded-range instances on the sparse random graph, the algorithm mis-eliminated the strongest item in most seeds — its boundary contrast must route through chains of saturated edges whose uncertainty the clipped plug-in radii understate. A principled one-sided treatment of saturated edges is the main open algorithmic problem. **(ii) The worst-case connectivity law is loose for identification.** E2b's five-point regression of $T^\star$ on $m_{\text{edges}}/\lambda_G$ has slope -0.06 against the predicted

1 – a noisy threshold-derived estimate that we read as indicative, not as a falsification – while E2c shows the law's bite once the boundary straddles a bottleneck (factor 3.92): identification cost is governed by boundary geometry, not worst-case connectivity alone. **(iii) The measured $\Delta$-dependence does not support Theorem 5.1's $\Delta^{-2}$ law at measurable gaps.** The exponent glides from -1.70 ($\Delta < 0.2$) to -0.55 ($\Delta > 0.6$), and we derive the deviation from the stopping rule: inverting the contrast threshold – whose Bernstein linear term supplies a $\Delta^{-1}$ component – yields a one-parameter law that fits the measured curve with $R^2 = 0.99$ in raw $T$ space, under the explicitly stated variance-scaling premise (Section 6.2); the one-shot estimator's $m^\star$ was too noisy to isolate its exponent. **(iv) Every error rate is measured** from recorded per-repetition outcomes in `results/*.json`; none is extrapolated.

# 7 Discussion

**Model assumptions.** The BT model is a parametric idealization; win probabilities need not be transitive, and the Borda/Copeland perspectives [27, 3, 30] address ranking functionals that remain meaningful when $\theta$ does not exist. Utilities are static; drifting preferences are out of scope. The bounded-range condition is standard [8, 33] but remains an assumption, and our practical radii use it as a clip rather than deriving it.

**What is proved and what is calibrated.** Theorem 5.1 is proved (by reduction to [33, 19]) for the one-shot estimator; Algorithm 1 is a calibrated heuristic (appendix) that achieved a 100% empirical success rate on the complete graph at every E1/E3 configuration but failed on sparse graphs with unbounded-range utilities (Section 6.7) — precisely the regime the bounded-range condition excludes. A rigorous sequential analysis of Algorithm 1 remains open.

**Adaptivity and the $\lambda_G$ penalty.** Our lower bound covers edge-balanced algorithms; Remark 4.3 explains why the fully adaptive rate needs a packing argument we could not complete, and E2 shows adaptive concentration reducing but not eliminating the connectivity penalty on the families tested. Fixed-budget formulations [13] and the semi-random graphs of [32] are natural next steps.

# 8 Conclusion

We formulated top-$k$ identification from noisy pairwise comparisons under a fixed, restricted comparison graph and gave, to the best of our knowledge, its first $\lambda_G$-explicit sample-complexity analysis: a one-shot uniform-per-edge estimator achieves $O(m_{\text{edges}} \log(n/\delta)/(\sigma_{\min}\lambda_G\Delta^2))$ comparisons, edge-balanced algorithms need $\Omega(m_{\text{edges}} \log(1/\delta)/(n\lambda_G\Delta^2))$, and both degenerate correctly to the classical complete-graph rates. Experiments validate the $n$-scaling, show a gap-dependence that we derive from the stopping rule, softer than the $\Delta^{-2}$ benchmark, expose the connectivity penalty when the boundary straddles a bottleneck, and reveal a sharp failure of naive baselines on sparse graphs. The two open fronts are the fully adaptive lower bound (Conjecture 4.2) and a rigorous sequential analysis of the practical algorithm.

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